Bihar Engineering University, Patna

B.Tech 6th Semester Examination, 2025

Course: B.Tech | *Time: 03 Hours*
Code: 155604 | *Subject: AI and Ethics* | *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
(iv) Question No. 1 is compulsory.

Q.1 ***Choose the correct option / answer the following (Any seven question only):*** **[2 x 7 = 14]**

(a) Morality in the context of AI primarily refers to:
- (i) Legal constraints on AI usage
- (ii) Principles that guide right and wrong behaviour in AI systems
- (iii) Technical limitations of AI algorithms
- (iv) Business rules followed by AI companies

(b) Which of the following is a potential negative psychological impact of AI on humans?
- (i) Increased emotional resilience
- (ii) Overdependence on automation
- (iii) Reduced digital interactions
- (iv) Improved mental health

(c) AI's impact on the legal system includes:
- (i) Removing all need for human judges
- (ii) Challenges in assigning responsibility for autonomous decisions
- (iii) Eliminating crime prediction tasks
- (iv) Making courts fully automated

(d) A major ethical concern regarding autonomous vehicles is:
- (i) Fuel efficiency
- (ii) Decision-making during unavoidable accidents
- (iii) Internet bandwidth usage
- (iv) Color of the car

(e) Healthcare robots raise ethical issues mainly related to:
- (i) Battery performance
- (ii) Patient privacy and safety
- (iii) Noise levels in hospitals
- (iv) Their manufacturing cost

(f) Which aspect of AI systems ensures users understand how the system makes decisions?
- (i) Classification accuracy
- (ii) Transparency in autonomous systems
- (iii) Dataset size
- (iv) Battery consumption

(g) Algorithmic bias refers to:
- (i) Faster execution of biased algorithms
- (ii) Systematic errors in AI that unfairly favour certain groups
- (iii) The speed at which an algorithm learns
- (iv) Errors caused by software bugs

(h) Which moral theory stresses consequences of actions when evaluating ethical decisions in robotics?
- (i) Deontology
- (ii) Virtue Ethics
- (iii) Consequentialism
- (iv) Intuitionism

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(i) Ethical issues in ICT society typically include:
(i) Data privacy and digital surveillance
(ii) Hardware optimization
(iii) Faster network speeds
(iv) Improved graphic processing

(j) A major ethical challenge of applying AI in medicine is:
(i) Lack of available computers
(ii) High electricity consumption
(iii) Limited number of doctors
(iv) Risk of incorrect automated diagnosis without adequate oversight

Q.2 (a) Define morality and ethics in the context of Artificial Intelligence. Explain why understanding moral principles is essential for designing responsible AI systems. [7]

(b) Discuss in detail the relationship between AI and society. Explain how AI influences social structures, human values, and cultural norms. [7]

Q.3 (a) Explain the various impacts of Artificial Intelligence on human psychology, including dependency, cognitive changes, and human–machine relationships. [7]

(b) Describe in detail the impact of AI on the legal system. Discuss challenges in accountability, liability, and regulation of autonomous decision-making. [7]

Q.4 (a) Discuss the impact of AI on the environment and the planet. Include energy consumption, e-waste, carbon footprint, and sustainability issues. [7]

(b) Using one detailed case study explain the ethical concerns and harms associated with AI deployment. [7]

Q.5 (a) Explain major international ethical initiatives in AI. Discuss guidelines by EU, IEEE, and other global bodies that promote responsible AI. [7]

(b) Discuss different types of ethical harms in AI systems and explain how they affect individuals, organizations, and society. [7]

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Q.6 (a) Describe the model process for addressing ethical concerns during AI system design. Explain each stage with examples. [7]

(b) What is algorithmic bias? Explain different sources of bias in AI systems and discuss methods to identify and mitigate them. [7]

Q.7 (a) Explain the social and ethical implications of robotics. Discuss issues related to autonomy, human–robot interaction, and professional responsibility. [7]

(b) Explain major moral theories (Deontology, Utilitarianism, Virtue Ethics). Discuss how these theories apply to evaluating robotic behaviour or automated systems. [7]

Q.8 (a) Discuss the ethical issues involved in applying AI in medicine, including diagnostic systems, decision support, and patient privacy. [7]

(b) Explain national and international AI strategies (India, EU, USA, OECD, etc.). Discuss how these strategies promote safe, transparent, and human-centric AI. [7]

Q.9 (a) Discuss the importance of transparency and accountability in AI systems. Explain how lack of transparency can affect public trust, ethical decision-making, and fairness in automated systems. Provide relevant examples. [7]

(b) Explain in simple terms why countries create rules and guidelines for the safe and responsible use of AI. Describe any two basic ethical concerns that these rules try to address. [7]

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